Capture-admission sensitivity study

Controlled sensitivity analysis and reproduction checks

26 October 2026

Status. Selected results are included in thesis Section 6.4.4; this report retains the complete study.

Question and design

When does a quorum improve contextual admission, and when does that advantage disappear? The frozen grid has 278 parameter conditions, each evaluated with 100,000 independent synthetic snapshot trials for object-present and object-absent context. The same latent random inputs are reused across rules and conditions for paired comparison; they are not 55.6 million independent scenes.

The four witnesses retain the supplied visual profile: placement radius 20 m, range limit 40 m, height 3 m, and prover height 1.5 m. Range and RF observations use the inspected source equations and retained RF correction. Sensor errors and missing observations are declared hypothetical inputs. There is no camera, image recognition, measured error calibration, rendered occlusion, physical relay attack, temporal process, or network evaluation.

The single-witness reference selects one witness uniformly before each trial's observations. "Any one of four" means a positive report from any eligible witness, a different and more permissive rule. Every rule sees the same range, RF and contextual samples.

Reference result: independent 10% errors

Each witness has false-positive and missed-detection probabilities 0.1, with no missing observations or dishonest witnesses. Entries are admission percentages with pointwise 95% Wilson intervals. The latter quantify Monte Carlo uncertainty within this model, not uncertainty about real sensors.

Rule	Valid context (%)	Invalid context (%)
One preselected	89.89 [89.70, 90.07]	10.112 [9.927, 10.300]
Any one of four	99.99 [99.98, 99.99]	34.723 [34.429, 35.019]
2 of 4	99.62 [99.58, 99.66]	5.333 [5.195, 5.474]
3 of 4	94.67 [94.53, 94.80]	0.380 [0.344, 0.420]
4 of 4	65.27 [64.97, 65.56]	0.010 [0.005, 0.018]

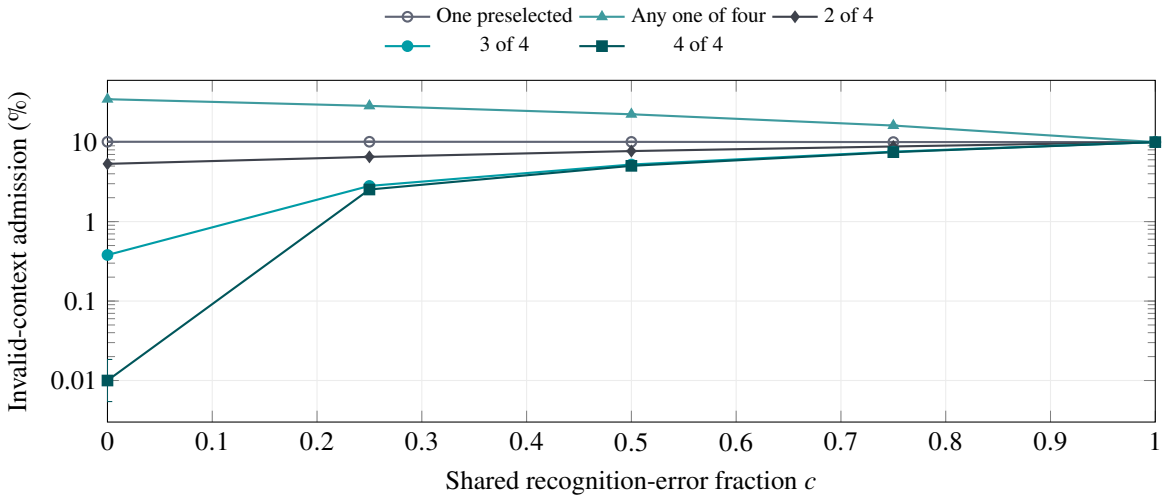
The range-plus-RF baseline admits 100.00% of the invalid-context trials: proximity and RF do not evaluate whether the object is present. The 3-of-4 contextual policy admits 94.67% of valid-context trials and 0.38% of invalid-context trials. These are conditional model results, not measured recognition or authentication performance.

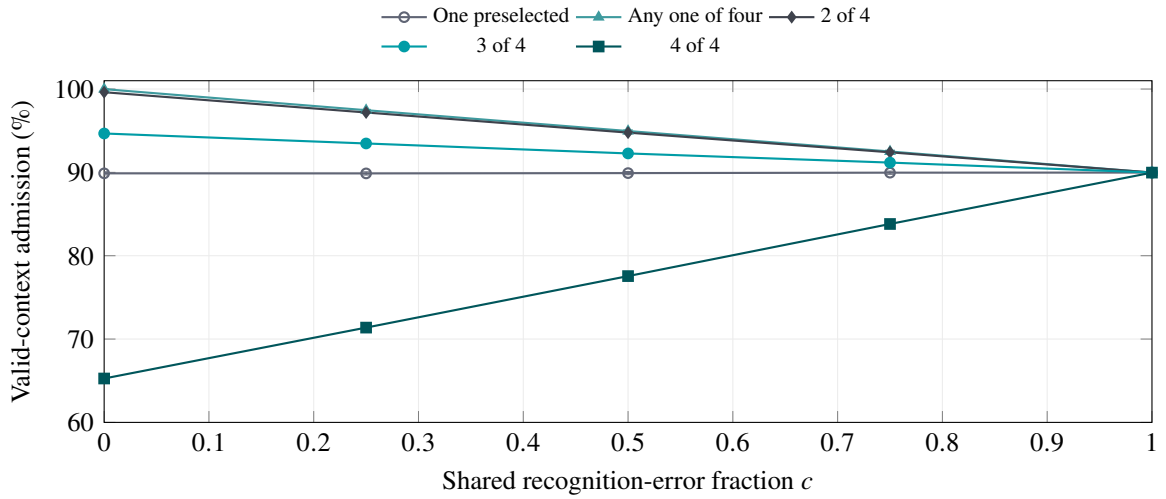
From the thesis root, with NumPy 2.5.2, SimPy 4.1.1, Python 3.12 and an authorised source checkout:

```
PYTHONHASHSEED=0 python3 \
 internal-review/experiments/capture-sensitivity/run-study.py \
 --source /path/to/ns3-sim-pol --output /tmp/sensitivity
```

Shared recognition errors

With probability c, all witnesses use one shared recognition-error draw; otherwise each uses an independent draw. Marginal error probabilities stay at 0.1. For homogeneous witnesses the raw pairwise error correlation is c; range and RF gating can introduce other decision dependencies. Other joint error distributions with the same pairwise correlation can produce different quorum rates.



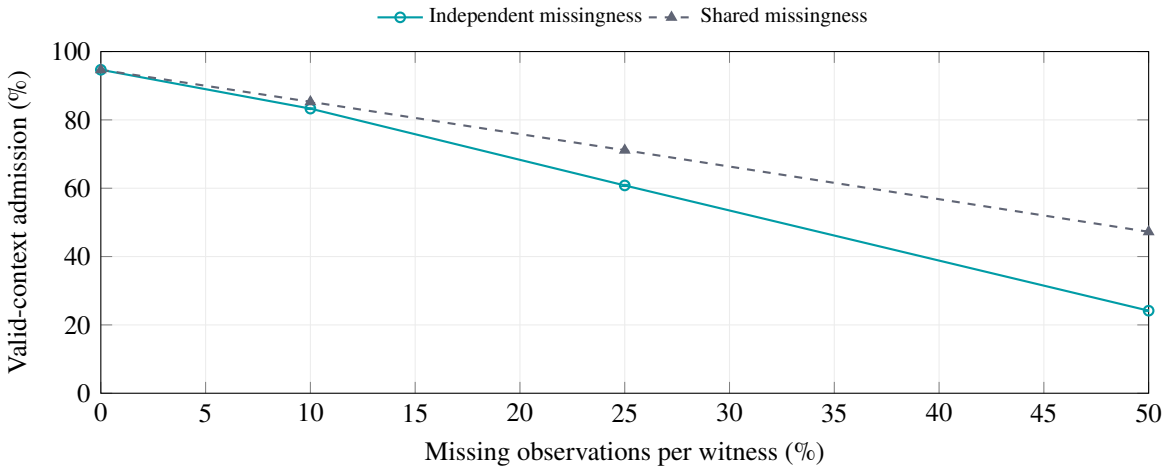


For 3-of-4, invalid-context admission rises from 0.38% to 10.02% as errors become fully shared. At full sharing, several observations contain essentially one recognition decision. Correlation does not worsen every rule in every direction: it reduces the false-positive union for “any one”, and reduces the chance of at least one missed detection for “all four”. The full results retain these effects.

Unavailable observations and dishonest reports

An invisible or missing patch produces UNKNOWN. Admission remains fail-closed. We additionally label a trial inconclusive when unknown reports from otherwise eligible witnesses could change the decision; a definitive rejection cannot be rescued by those unknown reports.

The following 3-of-4 curves keep recognition errors independent at 0.1 and vary only the missing observation process. With shared missingness, the entire observation set is missing together; independent missingness can instead fragment the set. Equal marginal availability therefore does not imply equal quorum availability.





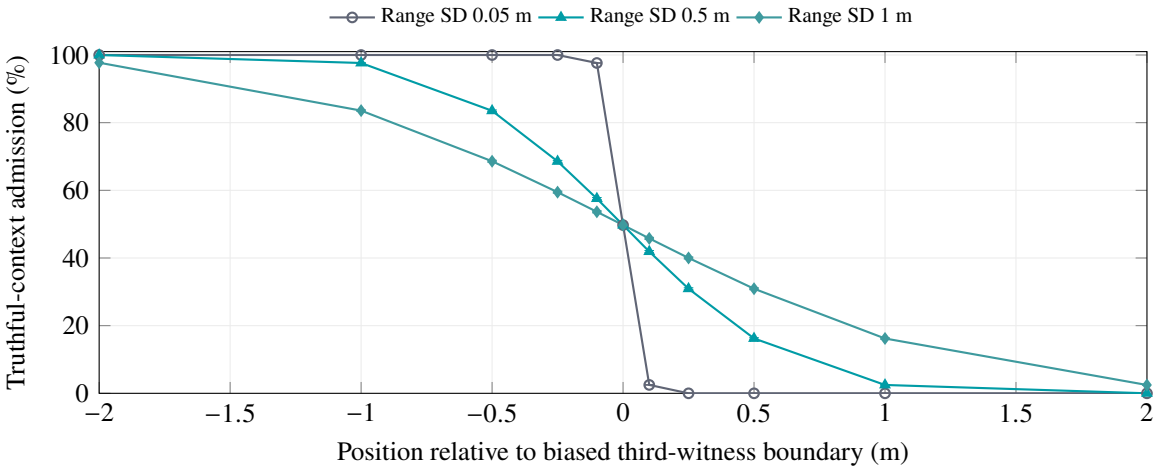
Dishonest witnesses. The first f witnesses always report positive. Ranges remain unforgeable within this model. The table gives invalid-context admission (%) for 3-of-4.

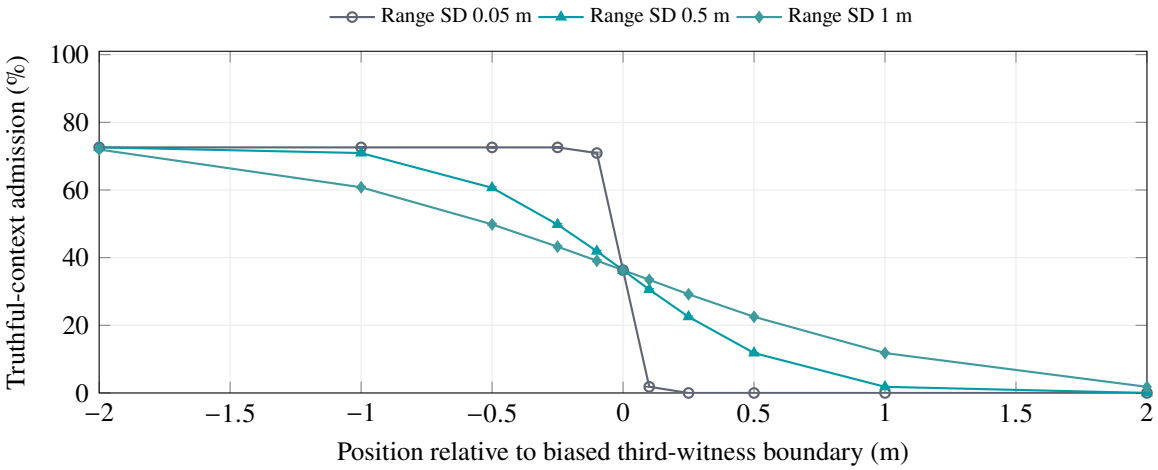
f	Independent errors	Half shared	Fully shared
0	0.38	5.23	10.02
1	2.81	6.48	10.02
2	19.31	14.69	10.02

With two dishonest positives, one further false positive suffices. Independence then creates two opportunities for that error, explaining why shared errors can lower admission in this particular compromised configuration. This is not a Byzantine-security proof.

Coverage boundary and range uncertainty

The path is $y = 5$ m. Zero on the horizontal axis is where the third witness's mean biased range equals 40 m: $x = 24.2612$ m. The lower panel adds independent 0.1 recognition error; the upper panel isolates ranging with perfect contextual observations. The unbiased geometric 40 m contour lies a further 0.509 m along this path.





These curves describe admission of truthful contextual claims at those positions. Outside the configured service region, non-admission is expected and is not automatically a false rejection. Range uncertainty softens the boundary in both directions. Near a three-witness contour, there may be only three useful observers, so even independent missed detections reduce availability more than at the four-observer interior point.

Viewpoint and observability

Each witness points toward the zone centre. Visibility requires the queried ground-level patch to lie within both its 3-D visual range and horizontal field of view. This is geometry, not an image-rendering model. The table gives the number of visible witnesses for every specified patch.

	Visual range 20 m			Visual range 50 m		
Patch (x, y) m	60°	120°	360°	60°	120°	360°
(0, 0)	0	0	0	4	4	4
(5, 5)	1	1	1	4	4	4
(20, 5)	0	0	1	1	2	4
(24, 5)	0	0	1	1	2	4
(25, 5)	0	0	1	1	2	4
(30, 5)	0	0	1	1	2	4
(0, 30)	0	0	0	2	2	4
(0, 45)	0	0	0	0	0	2
(35, 35)	0	0	0	0	0	1

The zero entries at the centre with a 20 m visual range follow from 3-D distance: ground-to-witness separation is $\sqrt{20^2 + 3^2} > 20$ m. They are a declared threshold effect, not evidence of a defective camera. At patch $(20, 5)$ with a 120° field of view and 50 m visual range, only 2 witnesses see the patch. The 3-of-4 valid-context admission is 0.00%. Witness placement and orientation therefore require coverage planning beyond the radio-proximity condition.

What the experiment establishes

The policy exhibits quantifiable trade-offs between invalid-context admission, valid-context admission, and observability under stated assumptions. Independent observations can help a quorum; shared errors, missing viewpoints, and dishonest reports can remove that benefit. Increasing the threshold can suppress invalid admission while reducing availability. These are conditional sensitivity findings with explicit failure conditions.

The visual predicate only asks whether a red car is present in a patch. It never inspects a submitted asset. This study cannot establish that a photograph depicts the observed object or was captured there; asset binding and physical sensing remain separate validation obligations.

Checks and reproducibility

All 2780 conditional analytical checks passed. The maximum deviation from the conditional analytical expectation was 3.54 Monte Carlo standard errors, within the predefined diagnostic tolerance of seven. This is an implementation check, not a goodness-of-fit test for real sensors. All 5760 sampled source-policy/quorum replays and 1668 nested-quorum checks passed. The corrected RF score passed its regression checks.

The complete grid, every result row, paired contrasts, and 100 batch counts per row are retained in `results/`. The design is `design.json`; the manifest records source, correction, driver, design, random-input hashes and dependency versions. Pointwise intervals follow the NIST Wilson-interval description. The inputs are independent synthetic snapshots, with no inferred real-world confidence bound. A second execution reproduced all five raw output files byte for byte. Before integration, hash checks confirmed that all 31 protected thesis and response files, including both PDFs, were unchanged. That pre-integration record is retained as historical provenance. The scripts `verify-study.py` and `make-report.py` check the repeat and generate this report.